



Game Theory: The Mekong River Water Conflict

Shreeniket Bhat



What are Water Conflicts?

Water conflicts arise where neighboring countries or states that share a transboundary water resource have opposing interests for the use of the water.

- Nile River: Egypt / Ethiopia
- Indus River: India / Pakistan
- Amu Darya River: Uzbekistan, Turkmenistan, Tajikistan and Afghanistan



The Mekong River

The Mekong River is one of the longest rivers in the world and crosses through China, Cambodia, Myanmar, Thailand, Vietnam, and Laos.

The Mekong River faces several issues including overfishing, plastic pollution, and effects of climate change, but I focused on the expansion of dam building.

The state of the Mekong has an impact on the livelihood of the people living in these Southeast Asian countries that rely on agriculture and fishing.



Dam Building – Pros and Cons



Pros:



- Electrical (renewable) energy generation



- Economic impact



Strategic reserves for:



- Irrigation



- Drinking water



- Water storage



- Flood Control

Cons:

- Affects flow rate, river path and biodiversity
- Floods and droughts
- Impact on livelihoods of people in upstream and downstream states
- Leaves downstream countries with insufficient water resources (e.g. blocks sediment flow)
- Long-term ecological impact
- Dam failures can be hazardous

Background

China, the upstream state, has built at least 11 hydropower mega dams on the mainstream and have built many more on tributaries leading into the river.

In retaliation to China's continuous expansion of dam projects, some of the downstream states, like Laos, have begun or have already created their own dam projects in the Mekong River.

For the downstream countries, the dams built by China seem to be the root cause of all their water resource problems, including recurring floods and droughts, despite China also having faced some of these same problems; the tensions still remain.



The Mekong River Commission

The MRC is an organization that works directly with the governments of Thailand, Cambodia, Laos, and Vietnam, the four main downstream countries, to effectively manage the shared water resources and the sustainable development of the Mekong River. The MRC was a significant player in the “Mekong River Game”.

They prioritize:

1. Protection of the Ecological Function of The Mekong (water flow and quality, sediment transport, ecosystem)
2. Optimal and Sustainable Development (sectoral and economic growth)
3. Resilience against Climate Risks (disaster management and preparation)
4. Inclusive Access/Utilization of Water Resources (poverty reduction and community well-being)
5. Strengthen Cooperation Among All Stakeholders/Countries (joint efforts, partnerships, projects, and agreements).



The Mekong River Game

I set this up as a two-player game with player 1 being China and the downstream countries collectively as player 2.

- Non-zero sum
- Asymmetric
- Sequential (China first)
- Mimics tragedy of the commons, but 2-player instead of n-player.



Strategies - China

Defect:

A) to continue constructing dams to boost renewable energy capacity

Cooperative:

B) to be willing to negotiate on a joint limit on the number of dams, while not joining MRC (middle-man).

C) to join the Mekong River Commission and provide support to downstream countries



Strategies – Downstream Countries

Defect:

- 1) to take a principled stance to stop constructing dams, and pressure China to join the MRC
- 2) build their own dams

Cooperative:

- 3) to accept whatever decision made by China, in the interest of trade and investment from China.



Utility Functions

It was difficult to obtain accurate and specific economic metrics on the Mekong River Conflict.

So based on an article which estimated the economic impact of dam construction in the Mekong, I generated hypothetical estimates of the costs and benefits to the two players.

I also considered the benefit of the completion of an FDI (investment) agreement, which is currently being discussed between the countries. This would only apply where the downstream countries play strategy 3.

1

		Downstream Countries			
		1	2	3	
China	A	(200, 0)	(250, 20)	(350, 220)	Stable solution; equilibrium point
	B	(120, 0)	(170, 40)	(270, 240)	Dominated by A
	C	X	(170, 40)	(180, 260)	Dominated by A and B

1. Dominated by 2 and 3
2. Dominated by 3

(Payoff Matrix 1 - China vs Others)

2

		Downstream Countries			
		1	2	3	
China	A	(200, 0)	(250, 20)	(350, 220)	Stable solution; equilibrium point
	B	(200, 0)	(250, 20)	(270, 240)	Dominated by A
	C	X	(250, 20)	(180, 260)	Dominated by A and B

1. Dominated by 2 and 3
2. Dominated by 3

(Payoff Matrix 2 - In Event of Rejected Agreement, Default Strategy for China is A)

3

		Downstream Countries			
		1	2	3	
China	A	(200, 0)	(250, 20)	(350, 220)	Dominated by B
	B	(120, 0)	(170, 40)	(370, 240)	Stable and Cooperative Solution; equilibrium point
	C	X	(170, 40)	(280, 260)	Dominated by A and B

1. Dominated by 2 and 3
2. Dominated by 3

(Payoff Matrix 3 - Considering China's Soft Power)

4

		Downstream Countries			
		1	2	3	
China	A	(200, 0)	(250, 20)	(350, 220)	Dominated by A and B
	B	(120, 0)	(170, 40)	(420, 390)	Stable and cooperative solution; equilibrium point
	C	X	(170, 40)	(330, 410)	Dominated by B

1. Dominated by 2 and 3
2. Dominated by 3

(Payoff Matrix 4 - Considering Long-Term Benefit of River Preservation and Sustainable Use of Resources)

Game Variations – Differences

Variation 1: This is the original with assumption that payoffs for B-1, B-2 and C-2, where agreements were rejected, follow same no. of dams if agreement was formed

Variation 2: the payoffs for B-1, B-2 and C-2 follow that of strategy A's number of dams.

Variation 3: Considers soft power or positive perception China gains in Southeast Asia when it plays a generous role and can conclude a cooperative outcome in B-3 as well as in C-3.

Variation 4: Considers long-term benefit of both players preserving the river in cooperative outcomes B-3 and C-3.

Game Variations – Common Conclusions

- Not fair game; biased toward China.
- China holds all the power
- Downstream countries have much larger variation in payoff; rely on China's cooperation, so have more to lose or gain.
- If the self-interested countries don't focus on short-term personal gain, they can work cooperatively to maximize their utilities within the constraints for greater long-term and collective gain. This reflects how the sheep herders should think in the tragedy of the commons.

Game Variations – Specific Conclusions

Variation 1 & 2:

- The solution to the game is outcome A-3 with payoffs (350, 220).
- Changing assumption for payoff values B-1, B-2, and C-2 had no effect on the solution in both variations.

Reflects reality of current situation:

- China unapologetically builds dams due to high economic/energy benefits
- Downstream countries not willing to retaliate because they have more to lose than China (e.g. trade or investment).

Variation 3 & 4:

- The solution to the games are outcome B-3 and payoffs (370, 240) and (420, 390), respectively.
- China does not consider its soft power from cooperation since this is not the current state of the situation.
- Countries may not consider long term benefit of keeping river preserved
- Ideal outcome for situation
- Being cooperative may not have greatest utility in short-run but in long-run it will prove beneficial.



**Thank you! Any
Questions?**

